

Whack-A-Drone

I. Scenario Description

A modified allied drone with a stick mounted onto its body has to navigate and hit the enemy drone. The drone's aim is to maneuver the stick to physically obstruct or disrupt the enemy drone's rotors, thus effectively disabling the enemy drone without resorting to more destructive methods such as shooting it down.

This scenario is important as it gives an approach to not only ensure the physical safety of human operators but also reduce the risk of accidental damage to bystanders or surrounding infrastructure. Moreover, this scenario offers a cost-effective solution for anti-drone warfare. The modification of existing allied drones to incorporate a stick is relatively affordable compared to developing new counter-drone technologies or deploying expensive interception systems. Additionally, this method avoids the need for ammunition or weapon systems, eliminating the associated costs of procurement, maintenance, and potentially harmful environmental impacts.

During this scenario, it is important for the allied drone to maintain a balance between disabling the enemy drone and taking an acceptable amount of damage. A careful assessment of the risk-benefit ratio will be made to determine the optimal approach. This ensures that the allied drone effectively accomplishes the mission while minimizing potential damage, allowing the drone to be used in subsequent operations.

II. Entities in the scenario

- Allied drone
- Enemy drone
- Surrounding infrastructures

III. States of the scenario

- Starting state: Allied drone is fully functional and flying or on the ground. The enemy drone is fully functional and flying or on the ground.
- Goal state: Allied drone is still functional and enemy drone is not.

IV. Possible actions that can be performed in the scenario and rewards

- Allied drone's stick hits the enemy drone's rotors - Large positive reward
- Allied drone's stick hits the enemy drone's body - Small positive reward
- Allied drone's body hits the enemy drone's rotors - Small positive reward
- Allied drone's body hits the enemy drone's body - Small negative reward
- Allied drone's rotors hits the enemy drone - Large negative reward
- Enemy drone hits the surrounding infrastructures - Small positive reward
- Allied drone hits the surrounding infrastructures - Large negative reward

V. Possible variations to the scenario

- The enemy is also a modified drone with a stick mounted onto it.
- There are multiple enemy drones.